

Ours-Direct 推荐层 × 真实编辑性能

Ours-Direct · M_abscos_x_newn = abs(S_v_cos) × S_v_new_norm | 7 模型 × 3 数据集 | 行内相对配色 · 格内仍为真实分数

模型 / 数据集		0–4					5–9					10–14					15–19					20–24					25–29					30–34					35	行内 相对值
		0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	
BLIP2 32 layers	E	58.16	58.51	50.13	62.46	57.26	68.89	—	—	—	—	73.09	—	—	—	75.19	73.04	73.80	75.60 ★	72.20	74.85	72.88	71.33	—	—	63.58	70.78	57.54	—	—	50.44	53.64	—					
	V	62.37	69.69	69.87	68.74	69.19	—	—	—	—	—	—	—	—	—	70.07	71.39	70.97	71.22	71.29	71.37	71.81	71.73	72.12 ★	—	71.36	71.47	71.08	—	—	69.30	65.77	—					
	N	65.86	72.50 ★	71.32	71.01	72.24	—	—	—	—	—	—	—	—	69.91	70.06	69.79	69.45	70.26	70.12	70.39	70.63	69.70	70.55	70.62	70.80	70.76	69.91	—	—	70.92	71.14	—					
InstructBLIP 32 layers	E	52.25	54.47 ★	49.95	50.47	47.88	—	—	—	—	49.24	49.91 ^e	49.46 ^e	49.90 ^e	—	44.86	50.22 ^e	—	—	—	—	—	—	—	—	51.70	51.44	52.27	53.27	—	—	—	—	—				
	V	71.29 ★	70.28	50.68	51.16	49.86	49.70	—	—	—	—	—	—	—	—	50.02	49.99	50.56	50.73	50.49	50.63	—	—	50.46	50.48	50.59	50.01	49.97	50.59	49.97	—	46.75	40.02					
	N	68.32	69.18 ★	49.79	50.02	48.89	48.20	—	—	—	—	—	—	—	—	48.64	48.84	48.72	48.93	48.62	48.60	—	—	48.13	48.33	47.27	47.58	47.52	46.94	46.87	46.70	46.48	40.12					
MiniGPT-4 32 layers	E	57.83	71.10	63.50	—	—	—	67.06	69.03	59.69	77.07 ★	—	—	—	63.43	64.07	64.94	63.91	59.56	58.38	—	—	58.55	—	57.70	58.47	57.27	—	58.36	58.53	57.84	54.67						
	V	75.81	75.82	76.06	76.63	—	76.94 ★	—	—	76.59	76.58	76.63	76.83	—	—	76.35	75.87	76.10	75.34	75.18	—	—	—	—	—	74.26	74.32	73.93	74.19	74.44	73.27	—	68.16					
	N	75.34	75.16	75.55	75.41	75.94	—	75.23	—	—	—	—	—	—	—	76.24	75.71	75.85	—	—	—	—	—	74.98	75.03	75.04	75.18	75.52	75.82	76.04	—	76.27 ★						
LLaVA-1.5 32 layers	E	—	—	—	—	—	70.09 ★	68.60	57.82	—	—	—	—	—	59.60	60.63	64.92	—	—	—	—	—	—	—	—	62.42	60.89	62.99	62.44	60.85	—	57.24	58.08					
	V	75.16	75.64	76.79	76.19	—	—	76.55	77.01 ★	76.91	—	—	76.07	—	76.51	75.75	75.74	—	—	—	—	—	—	73.84	—	74.59	—	73.42	74.40	73.53	—	—	—					
	N	—	—	—	—	—	—	—	—	—	—	—	—	—	76.03 ★	75.74	76.01	—	—	—	—	—	—	75.54	—	—	75.14	75.15	74.74	—	—	—						
Qwen2.5-VL 36 layers	E	62.89	61.46	62.25	61.59	—	—	—	—	—	—	61.83	62.67	—	63.39	62.40	62.76	62.60	63.55	63.21	62.11	63.74 ★	62.82	—	62.79	—	62.33	61.94	62.53	62.37	62.52	63.05	—	—	62.77	61.91		
	V	70.42	69.48	69.93	—	—	—	70.15	70.47	70.84	69.48	—	71.00 ★	71.00 ★	70.39	69.82	—	69.81	69.60	70.65	69.93	70.40	—	70.34	—	—	—	68.26	68.12	68.35	—	67.05	—	—	—	—		
	N	72.22	72.28	72.04	—	—	—	—	—	—	—	—	—	—	71.44	71.48	71.12	71.59	71.48	71.20	—	71.21	70.66	71.30	—	—	71.70	70.66	71.34	71.81	72.66 ★	72.41	—	—	—	—		
PaliGemma 18 layers	E	—	38.82	—	85.72 ★	71.13	78.17	66.67	75.65	81.00	36.16 [!]	70.52	80.99	75.98	—	32.06 [!]	40.11	—	40.11																			
	V	—	98.10 ^s	98.47 ^s	97.90 ^s	99.03 ^s	98.58 ^s	97.97 ^s	98.40 ^s	91.29	98.09	99.06 ★	97.07	96.88	91.53 ^s	39.03	—	—	40.05																			
	N	20.00 [!]	88.86	82.04	92.97 ^s	96.14 ^s	90.81 ^s	96.98 ^s	90.12	96.30	85.41	38.28	44.62	90.67	48.65	—	—	47.04 [!]	40.33																			
SmoIVLM 24 layers	E	69.97 ★	58.77	64.41	66.56	—	65.36	—	68.45	63.98	68.06	69.81	67.24	61.93	59.45	68.22	59.74	58.90	58.85	—	57.31	57.15	56.37	55.87	—													
	V	71.36 ★	70.18	70.52	—	70.53	70.32	69.88	70.77	69.66	69.45	69.34	69.62	68.52	69.83	68.64	68.48	67.19	67.25	67.89	67.34	67.40	67.16	—	—													
	N	70.47	71.05	71.12	—	—	—	70.49	70.30	—	70.44	70.62	69.55	70.84	69.65	69.32	69.53	69.59	70.59	71.47 ★	—	—	—	—	—													

越深越好

行内归一化

红色实框：本方法 Top-3

橙色虚框：本方法 第 4–5 名

角标 1–5：推荐次序

★：本行已测主配置最高分

—：文档无评测值

无此层

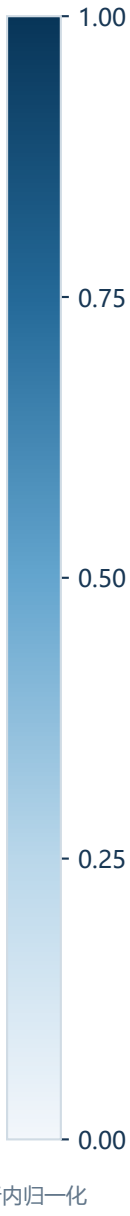
E = EVQA-pilot500 V = MMKE-visual N = MMKE-entity | Average = (Rel + T-Gen + M-Gen + T-Loc + M-Loc) / 5

配置：优先 main；s = 仅有 stable，e = 提前结束训练，! = 数值异常或恢复记录。★ 排除 stable 与 recovered-early；原始低分如实保留。

未评测不填 0；BLIP2-E 的 L18 使用原始记录，重复实验 L18-2 单列于数据文件。

颜色=(本行分数–本行最小值)/(本行最大值–本行最小值)，用于观察同组层间差异；不同组的深浅不可直接比较。

越深
越好



行内归一化